

Bad Banking Database

Query 1 x SQL File 3*

Limit to 1000 rows

1 • Select * from bad_bankingdb.credit_card_file;

Result Grid

	card_id	customer_id	customer_name	address	phone	card_number	card_type	
1	1	C001	Aung Aung	No.10, Pyay Road, Yangon	09-111111111	4111111111111111	Gold	2
2	2	C003	Kyaw Kyaw	No.45, Mandalay Road, Mandalay	09-333333333	4222222222222222	Classic	1
*	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

credit_card_files table

Query 1 x SQL File 3*

Limit to 1000 rows

1 • Select * from bad_bankingdb.account_file;

2

Result Grid

	account_id	customer_id	customer_name	address	phone	account_type	balance
1	1	C001	Aung Aung	No.10, Pyay Road, Yangon	09-111111111	Savings	500000.00
2	2	C002	Su Su	No.22, Hledan Street, Yangon	09-222222222	Current	1200000.00
3	3	C003	Kyaw Kyaw	No.45, Mandalay Road, Mandalay	09-333333333	Savings	300000.00
*	NULL	NULL	NULL	NULL	NULL	NULL	NULL

account_files table

Query 1 x SQL File 3*

Limit to 1000 rows

```
1 • Select * from bad_bankingdb.loan_file;
2
```

Result Grid

	loan_id	customer_id	customer_name	address	phone	loan_type	loan_amount	outstanding
▶	1	C001	Aung Aung	No.10, Pyay Road, Yangon	09-111111111	Home Loan	10000000.00	7500000.00
	2	C002	Su Su	No.22, Hledan Street, Yangon	09-222222222	Car Loan	5000000.00	2000000.00
*	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

loan_files table

=====

-- WORKFLOW PROBLEM

-- Customer C001 changes address

-- Only account_file is updated

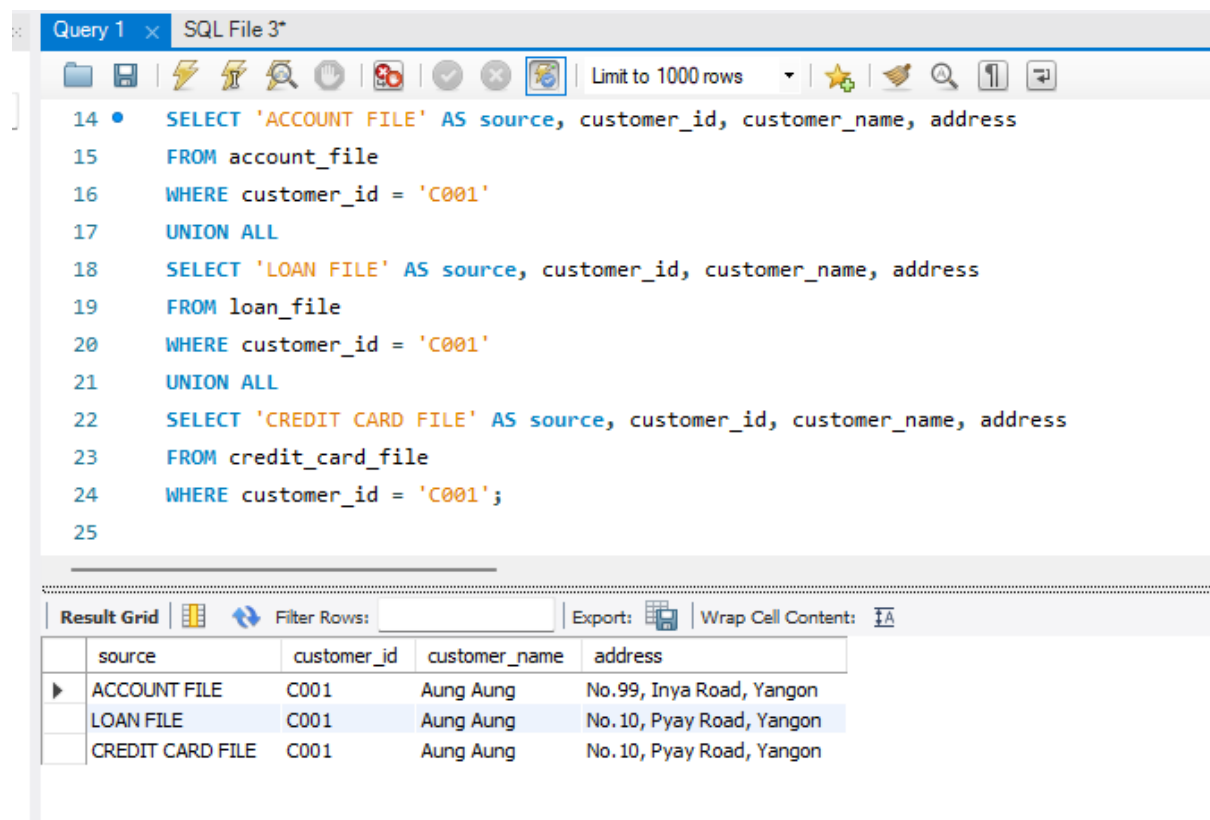
=====

UPDATE account_file

SET address = 'No.99, Inya Road, Yangon'

WHERE customer_id = 'C001';

After this update, let's check consistency



The screenshot shows a SQL query editor with a query window titled 'Query 1' and a file named 'SQL File 3*'. The query is as follows:

```
14 • SELECT 'ACCOUNT FILE' AS source, customer_id, customer_name, address
15 FROM account_file
16 WHERE customer_id = 'C001'
17 UNION ALL
18 SELECT 'LOAN FILE' AS source, customer_id, customer_name, address
19 FROM loan_file
20 WHERE customer_id = 'C001'
21 UNION ALL
22 SELECT 'CREDIT CARD FILE' AS source, customer_id, customer_name, address
23 FROM credit_card_file
24 WHERE customer_id = 'C001';
25
```

Below the query editor, the 'Result Grid' is displayed, showing the results of the query. The table has four columns: 'source', 'customer_id', 'customer_name', and 'address'. The results are as follows:

source	customer_id	customer_name	address
ACCOUNT FILE	C001	Aung Aung	No.99, Inya Road, Yangon
LOAN FILE	C001	Aung Aung	No.10, Pyay Road, Yangon
CREDIT CARD FILE	C001	Aung Aung	No.10, Pyay Road, Yangon

Inconsistency data

Good Banking Database

Query 1 x SQL File 3*

Limit to 1000 rows

```
1 select * from good_bankngdb.loans;
```

Result Grid Filter Rows: Edit: Export/Import: Wrap Cell C

	loan_id	customer_id	loan_type	loan_amount	outstanding_amount	start_date
▶	1	C001	Home Loan	10000000.00	7500000.00	2024-04-01
	2	C002	Car Loan	5000000.00	2000000.00	2024-05-10
*	NULL	NULL	NULL	NULL	NULL	NULL

loans table

Query 1 x SQL File 3*

Limit to 1000 rows

```
1 select * from good_bankngdb.accounts;
```

Result Grid Filter Rows: Edit: Export/Imp

	account_id	customer_id	account_type	balance	opened_date
▶	1	C001	Savings	500000.00	2024-01-10
	2	C002	Current	1200000.00	2024-02-15
	3	C003	Savings	300000.00	2024-03-01
*	NULL	NULL	NULL	NULL	NULL

accounts table

Query 1 x SQL File 3*

Limit to 1000 rows

```
1 select * from good_bankingdb.credit_cards;
```

Result Grid

	card_id	customer_id	card_number	card_type	credit_limit	expiry_date
▶	1	C001	4111111111111111	Gold	2000000.00	2028-12-31
	2	C003	4222222222222222	Classic	1000000.00	2027-11-30
*	NULL	NULL	NULL	NULL	NULL	NULL

credit_cards table

Query 1 x SQL File 3*

Limit to 1000 rows

```
1 select * from good_bankingdb.customers;
```

Result Grid

	customer_id	customer_name	address	phone	email	created_at
▶	C001	Aung Aung	No.10, Pyay Road, Yangon	09-111111111	aung@example.com	2026-04-25 19:03:24
	C002	Su Su	No.22, Hledan Street, Yangon	09-222222222	susu@example.com	2026-04-25 19:03:24
	C003	Kyaw Kyaw	No.45, Mandalay Road, Mandalay	09-333333333	kyaw@example.com	2026-04-25 19:03:24
*	NULL	NULL	NULL	NULL	NULL	NULL

customres table

UPDATE customers

SET address = 'No.99, Inya Road, Yangon'

WHERE customer_id = 'C001';

-- Check consistent result through joins

```
1 SELECT FROM g004_bankingdb.customers;
2 • SELECT
3     c.customer_id,
4     c.customer_name,
5     c.address,
6     a.account_type,
7     a.balance,
8     l.loan_type,
9     l.outstanding_amount,
10    cc.card_number,
11    cc.card_type
12 FROM customers c
13 LEFT JOIN accounts a ON c.customer_id = a.customer_id
14 LEFT JOIN loans l ON c.customer_id = l.customer_id
15 LEFT JOIN credit_cards cc ON c.customer_id = cc.customer_id
16 WHERE c.customer_id = 'C001';
```

Result Grid

Filter Rows:

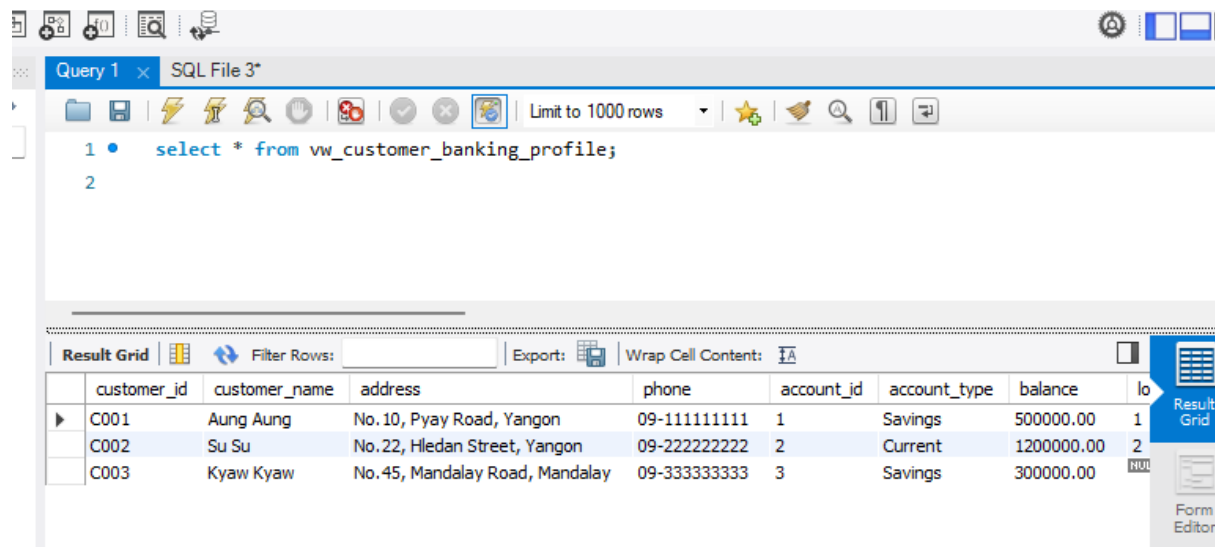
Export:

Wrap Cell Content:

	customer_id	customer_name	address	account_type	balance	loan_type	outstanding_amount	car
	C001	Aung Aung	No. 10, Pyay Road, Yangon	Savings	500000.00	Home Loan	7500000.00	411

consistency data

-- View for Easy Demonstration



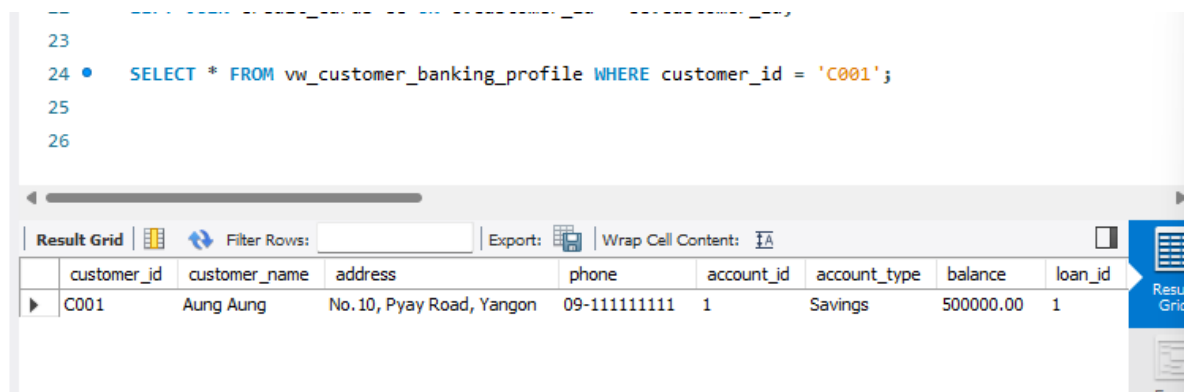
The screenshot shows a SQL IDE window titled "Query 1" and "SQL File 3*". The query editor contains the following SQL statement:

```
1 • select * from vw_customer_banking_profile;
2
```

The results are displayed in a "Result Grid" table with the following data:

	customer_id	customer_name	address	phone	account_id	account_type	balance	loan_id
▶	C001	Aung Aung	No.10, Pyay Road, Yangon	09-111111111	1	Savings	500000.00	1
	C002	Su Su	No.22, Hledan Street, Yangon	09-222222222	2	Current	1200000.00	2
	C003	Kyaw Kyaw	No.45, Mandalay Road, Mandalay	09-333333333	3	Savings	300000.00	3

View table



The screenshot shows a SQL IDE window with a query that filters the results by customer_id:

```
23
24 • SELECT * FROM vw_customer_banking_profile WHERE customer_id = 'C001';
25
26
```

The results are displayed in a "Result Grid" table with the following data:

	customer_id	customer_name	address	phone	account_id	account_type	balance	loan_id
▶	C001	Aung Aung	No.10, Pyay Road, Yangon	09-111111111	1	Savings	500000.00	1

Test Case

Test case (1) with bad_bankingdb

After updaing these (

UPDATE account_file

SET address = 'No.10, Pyay Road, Yangon'

WHERE customer_id = 'C001';

UPDATE loan_file

SET address = 'No.10, Pyay Road, Yangon'

WHERE customer_id = 'C001';

UPDATE credit_card_file

SET address = 'No.10, Pyay Road, Yangon'

WHERE customer_id = 'C001';

-- Update only account file

UPDATE account_file SET address = 'No.99, Inya Road, Yangon' WHERE
customer_id = 'C001';

)

Check consistency

```
277 -- Verify inconsistency
278 • SELECT 'ACCOUNT FILE' AS source, address
279 FROM account_file
280 WHERE customer_id = 'C001'
281 UNION ALL
282 SELECT 'LOAN FILE' AS source, address
283 FROM loan_file
284 WHERE customer_id = 'C001'
285 UNION ALL
286 SELECT 'CREDIT CARD FILE' AS source, address
287 FROM credit_card_file
288 WHERE customer_id = 'C001';
289
```

Result Grid			Filter Rows:	Export:	Wrap Cell Content:
	source	address			
▶	ACCOUNT FILE	No.99, Inya Road, Yangon			
	LOAN FILE	No. 10, Pyay Road, Yangon			
	CREDIT CARD FILE	No. 10, Pyay Road, Yangon			

Inconsistency Data

Test Case (2) with good_bankingdb

Run these (

-- Reset centralized data

```
UPDATE customers SET address = 'No.10, Pyay Road, Yangon' WHERE  
customer_id = 'C001';
```

```
-- Update once UPDATE customers SET address = 'No.99, Inya Road, Yangon'  
WHERE customer_id = 'C001';
```

)

-- Verify consistency

```
303 • SELECT  
304     c.customer_id,  
305     c.customer_name,  
306     c.address,  
307     a.account_type,  
308     l.loan_type,  
309     cc.card_type  
310 FROM customers c  
311 LEFT JOIN accounts a ON c.customer_id = a.customer_id  
312 LEFT JOIN loans l ON c.customer_id = l.customer_id  
313 LEFT JOIN credit_cards cc ON c.customer_id = cc.customer_id  
314 WHERE c.customer_id = 'C001';  
315
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: [IA](#)

	customer_id	customer_name	address	account_type	loan_type	card_type
▶	C001	Aung Aung	No.99, Inya Road, Yangon	Savings	Home Loan	Gold

Consistency Data